PAPER_255: PSR J0030+0451 Isolated Neutron Star — Density Regime Positive Buoyancy and F_neutron Dominance

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$$F_U(r,t) = \sum_{i=1}^{4} U_{gi} + U_m + U_A - U_{b_i}, \quad \kappa = 5.0 \times 10^{-4}\,\text{day}^{-1},\ [SSq] = 0.57$$

$$M_J^{\text{UQFF}} = M_J^{\text{Jeans}} \cdot \left(1 - [SSq] \cdot \frac{B^2}{8\pi\rho c_s^2}\right), \quad [SSq] = 0.57$$

Abstract

PSR J0030+0451 is an isolated millisecond pulsar at ~1,100 light-years, with a mass of approximately 1.4 M_sun confined to a radius of ~10 km (r = 104 m) — the compact geometry of a neutron star. This system is the **first isolated pulsar class** in the CP3 calculator, and it introduces a new UQFF regime defined by the neutron-star-density cross-section parameter `s_n ~ 10`^{3}?, representing the degenerate nuclear density of neutron star matter.

In the ISM systems of PAPER_250–254, s_n ~ 10¯4 yields F_neutron = k_neutron × s_n = 106 N. For PSR J0030 at s_n = 10^3?, F_neutron = 10^1° × 10^3? = **104? N** — a difference of 53 orders of magnitude. This neutron force is the dominant UQFF term by far.

The key **uniquely rare discovery** of this paper is that despite this 53-order amplification of F_neutron, and despite the compact scale (r = 104 m vs r = 6.17 × 10^{1}6 m for the SNRs), PSR J0030 is a **positive buoyancy** system: F_U_Bi ~ +2.53 × 10^2°8 N. The compact-scale geometry at ?0 = 10?^{12} preserves the positive sign. The equivalence class extends across 14 orders of magnitude in radius and 53 orders in s_n.

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1. System Parameters

Parameter	Symbol	Value	Units	Source
Distance	d	~1,100	ly	Chandra/NICER 2019
Mass	M	1.4 M_sun = 2.786 × 10^3°	kg	Neutron star canonical
Neutron star radius	**r**	**104**	**m**	**~10 km**
X-ray luminosity	L_X	10^{31}	W	NICER 2019
Surface B field	B0	108	T	Millisecond pulsar typical
System frequency	?0	10?^{12}	rad/s	Same as SNR class
Neutron cross-section	**s_n**	**10^3?**	—	**NS density (vs ISM 10?4)**

2. Core Physics: Neutron-Star Density Regime

2.1 F_neutron — The New Dominant Term

```
F_neutron = k_neutron × s_n = 10^1° × 10^3? = 104? N
```

For comparison:

```
F_LENR (?0=10?^12) ~ 6.17 × 10^3? N
F_neutron / F_LENR = 104? / 6.17×10^3? ~ 1.6 × 10?
```

F_neutron exceeds F_LENR by 9 orders for the neutron star regime. The force hierarchy shifts from LENR-dominant (ISM and SNR systems) to **neutron-dominant** (compact objects):

```
Force hierarchy at ?0=10?^12, s_n=10^3?:
F_neutron ~ 104? N   [dominant — 9 orders above F_LENR]
F_LENR   ~ 6×10^3? N   [second]
F_Newt   ~ GM/r^2·|x2| [negligible]
F_res    « F_LENR      [DPM invisible — same conclusion as PAPER_251]
```

2.2 Compact Geometry and P Positive Buoyancy Preservation

Despite the 9-order dominance of F_neutron over F_LENR, the sign of F_U_Bi remains positive. This is because the compact geometry (r = 104 m) affects the term_gravity = GM/r^2 and the integration limit x2 in a way that preserves the positive root:

```
term_gravity = G·M/r^2 = 6.674e-11 × 2.786e30 / (104)^2
             ~ 1.86 × 106 m/s^2   [huge surface gravity]
```

The quadratic discriminant `b^2 - 4ac` with `a = 1.86×106`, `b = 4.72×10?^3`, `c ~ -1.83×107^1` gives a positive x2 root (same sign as ISM systems), because the vacuum energy F0 = 1.83×107^1 N overwhelms the sign-determining coefficient c regardless of the surface gravity scale.

Key result: `x2 > 0 ? F_U_Bi_i = integrand × |x2| > 0 ?` **positive buoyancy at F_U_Bi ~ +2.53 × 10^2°8 N**.

2.3 53-Order s_n Range: Equivalence Class Breadth

The s_n parameter spans:

```
s_n (ISM/SNR systems):  ~ 10^-4 ? F_neutron = 106 N  [PAPER_250–254]
s_n (PSR J0030):        ~ 10^3? ? F_neutron = 104? N [this paper]
```

53 orders of magnitude in s_n, yet both classes show **positive buoyancy at ?0 = 10?^{12}**. This confirms that s_n (like B0 in PAPER_251) does not breach the equivalence class — the ?0 parameter remains the exclusive class determinant.

2.4 DPM Resonance at B0 = 108 T

```
DPM_resonance (PSR J0030) = 2·μ_B·B0/(h·?0)
                           = 2 × 9.274e-24 × 108 / (1.0546e-34 × 10?^12)
                           ~ 1.76 × 10^31
```

This is an astronomically large DPM resonance, yet it is still invisible relative to F_neutron: F_res/F_neutron « 1. The DPM invisibility theorem (PAPER_251) extends to the neutron-star-density regime.

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3. Extended Force Equivalence Class Theorem

Theorem (UQFF NS-Density Class Extension): The positive buoyancy equivalence class at ?0 = $10?^{12}$ rad/s includes compact objects with neutron-star densities (s_n ~ 10^{3}?) in addition to diffuse ISM systems (s_n ~ 10?4). F_U_Bi ~ +2 × 10^{2}°8 N regardless of s_n spanning 53 orders, confirming that s_n is not a class-determinant parameter. The vacuum energy anchor F0 = 1.83 × 107^{1} N ensures x2 > 0 for all physically observable values of s_n.

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4. ALMA Cycle 12 Observational Context

PSR J0030+0451 is an ALMA Cycle 12 proposal target. Observable UQFF signatures include:

- **Isotopic anomaly:** LENR neutron-capture at F_neutron = 104? N (53 orders above ISM) predicts elevated $^{2}H/^{1}H > 10^{-}5$ and $^{13}C/^{12}C > 0.01$ in the pulsar wind nebula — detectable with ALMA Band 6 at 230 GHz.
- **EHT polarimetry:** The extreme DPM_resonance ~ 1.76 × 10^{31} at B0 = 108 T predicts distinctive helical B-field structure in the pulsar wind, detectable with EHT 20 µas resolution at 230 GHz.
- **NICER hotspot:** PSR J0030+0451 hotspot morphology constrains the NS mass-radius relation; UQFF predicts F_U_Bi positive — consistent with a gravitationally stable bound NS (no anomalous mass loss or unbinding).

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5. References

1. Riley, T.E. et al. (2019). A NICER View of PSR J0030+0451: Millisecond Pulsar Parameter Estimation. *ApJ Lett.* 887, L21.
2. Özel, F., & Freire, P. (2016). Masses, Radii, and the Equation of State of Neutron Stars. *ARA&A* 54, 401.
3. Kozima, H. (1998). *The Science of the Cold Fusion Phenomenon*. Elsevier.
4. ALMA Proposal Cycle 12. PSR J0030+0451 — UQFF isotopic anomaly detection (Murphy, D.T. 2026).
5. Murphy, D.T. (2026). UQFF Framework v4.27 — NS Density Regime Discovery. Star-Magic Session 72d.

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